Åbo Akademi University

Systematic Mapping and Benchmarking of Sign Language Datasets

Author: Kaniz Fatema Begum
Supervisor: Luigia Petre

Content

1. Introduction

Sign languages are natural languages used by deaf and hard-of-hearing communities worldwide. In contrast to spoken languages, sign languages are visually conveyed through hand movements, facial expressions, and body posture, and have their own distinct linguistic structures. Recent advances in artificial intelligence and machine learning have increased research interest in automatic sign language recognition, translation, and accessibility technologies. Progress in these domains relies on the availability of suitable and well-documented datasets.

In the past decade, a wide range of sign language datasets have been introduced, covering various sign languages, recording environments, and research objectives. These datasets vary significantly in scale, modality (such as RGB video or depth data), annotation schemes, and signer diversity. However, information regarding these datasets is scattered across academic publications, project websites, and repositories. Consequently, researchers, particularly those new to the field, frequently encounter challenges in identifying suitable datasets and comprehending their limitations.

The lack of a consolidated and structured overview of sign language datasets hinders research progress and reproducibility. In the absence of clear comparisons and classifications, it is difficult to assess dataset usability, identify gaps in language coverage, or determine the suitability of datasets for specific research tasks. This challenge highlights the necessity for a systematic approach to mapping and evaluating existing sign language datasets.

This thesis aims to deliver a structured overview of existing sign language datasets by identifying, categorizing, and comparing their key characteristics. The analysis focuses on properties such as language coverage, modality, scale, annotation, and accessibility. By applying a systematic mapping methodology, the thesis clarifies the current landscape of sign language datasets and highlights existing limitations and research gaps.

2. Background and Related Work

This chapter establishes the conceptual and scholarly context for the systematic mapping study presented in this thesis. It introduces the key concepts, prior work, and methodological frameworks that inform the design and scope of this research.

2.1 Sign Language Datasets in Computational Research

Sign language datasets serve as the primary empirical foundation for computational sign language research. These datasets comprise video recordings of one or more signers, with each clip annotated to indicate the identity of the sign or signs being performed. They are utilized to train, evaluate, and benchmark models for a variety of tasks.

Existing datasets differ substantially in scope, including the sign language represented, the number of signs and signers, the type of data recorded, and the level of annotation detail. These variations reflect distinct research priorities and directly influence the tasks a dataset can support as well as the reliability of experimental results. Understanding these characteristics is essential for selecting appropriate datasets and for comparing results across studies.

2.2 Types of Sign Language Datasets

A primary distinction among datasets is the type of signing captured: isolated signs or continuous signing.

An isolated sign dataset consists of short video clips in which a single sign is performed independently. Each clip is labelled with the name of the sign, known as a gloss, which is a written word used to identify a sign (e.g., WATER, HELLO). These datasets support the task of sign language recognition (SLR), which involves identifying the sign being performed in a given video clip. A widely used example is WLASL (Word-Level American Sign Language), which contains over 21,000 clips covering 2,000 signs performed by 119 signers.

A continuous signing dataset captures naturally flowing sign language, where multiple signs are produced sequentially to form phrases and sentences. These datasets are more complex to collect and annotate because sign boundaries must be identified within the video stream. They support tasks such as sign spotting, which involves detecting where specific signs occur within a continuous signing stream, and sign language translation (SLT), which converts a signed video into a written or spoken sentence in another language. An example is PHOENIX-2014, a

German Sign Language dataset of weather forecast broadcasts containing approximately 7,000 sentences.

2.3 Data Modalities

The term modality refers to the type of sensor or data format used to record signing. Datasets may include one or multiple modalities, each capturing distinct aspects of the signing event.

RGB video, which is standard color video from a conventional camera, is the most common modality. It captures the full visual appearance of signing but is sensitive to lighting conditions and background clutter. Depth data, recorded with sensors such as the Microsoft Kinect, provides distance information for each pixel, enabling more robust foreground segmentation and three-dimensional analysis, although it requires specialized hardware.

Skeleton or pose data represents the signer as a set of body joint coordinates, such as wrist, elbow, and finger positions. These coordinates may be estimated automatically from RGB video using tools such as MediaPipe or captured directly via motion-capture equipment. Pose data is compact and less sensitive to visual factors such as clothing; however, automatic estimation can be unreliable for rapid hand movements or when one hand obscures the other. Some datasets record signing simultaneously from multiple camera angles (multi-view), which reduces occlusion but significantly increases the complexity of data collection.

2.4 Annotation in Sign Language Datasets

Annotation refers to the descriptive labels assigned to raw video data. The depth of annotation determines the range of tasks a dataset can support.

The most common form of annotation is the gloss label, which assigns a sign name to each clip or segment. In continuous datasets, annotations must also specify the start and end points of each sign, a process known as temporal segmentation, which is necessary for sign spotting tasks. Datasets intended for SLT additionally provide a spoken-language translation of each signed phrase, pairing signed video with a corresponding written sentence (for example, PHOENIX-2014-T includes German sentence translations alongside gloss labels). A small number of datasets include phonological annotations that describe the sub-lexical structure of signs, such as hand shape, location, and movement, primarily for linguistic analysis rather than recognition.

2.5 Dataset Scale and Signer Diversity

The scale of a dataset is typically measured by vocabulary size (the number of distinct sign classes), the total number of video samples, and the number of signers. These dimensions collectively determine the extent of variation within the dataset and the degree to which a trained model can generalize to unseen signing.

A larger vocabulary and a greater number of samples per sign generally support more robust model training. The number of signers is particularly important; a dataset with only five signers exposes the model to a limited range of signing styles, which can artificially inflate performance when tested on the same individuals. Datasets with more signers, such as AUTSL (43 signers) or WLASL (119 signers), offer greater diversity, although they remain limited compared to the full breadth of a signing community.

Demographic factors such as age, gender, handedness, and whether signers are deaf native users or hearing learners also influence the representativeness of a dataset. This information is often absent or inconsistently reported, which complicates the assessment of potential bias in trained models.

2.6 Benchmarking and Evaluation Protocols

Benchmarking involves evaluating a model on a standardized test to facilitate comparison across studies. The evaluation protocol, which defines how the dataset is divided into training and test portions, has a significant impact on reported performance and its practical interpretation.

In a signer-dependent evaluation, some signers appear in both the training and test sets, meaning the model has already been exposed to each test signer's style, which can inflate accuracy. In contrast, a signer-independent evaluation ensures that test signers are entirely absent from training, providing a more realistic measure of generalization. Signer-independent protocols are widely regarded as the more rigorous standard but require datasets with a sufficient number of signers to allow for separation across splits.

When a fixed train, validation, and test split is published alongside the dataset as a standardized benchmark, all research groups can evaluate under identical conditions, making results directly comparable. In the absence of such a standard, different groups may partition the same data differently, resulting in outcomes that are difficult to compare. Performance is typically reported using metrics such as Top-1 accuracy for isolated recognition or Word Error Rate (WER) for continuous tasks.

2.7 Dataset Accessibility and Licensing

Accessibility refers to whether a dataset is publicly available, whether it requires a formal application, and the licensing terms that govern its use. Many sign language datasets are distributed under research-only licenses; some are open, while others are restricted to academic institutions. A small number have become unavailable over time due to institutional changes or data protection regulations.

The quality of documentation also affects practical usability. A well-documented dataset includes descriptions of the collection procedure, signer demographics, annotation methodology, file formats, and baseline evaluation results. Inconsistent or incomplete documentation hinders the reproducibility of experiments and reduces a dataset's adoption within the research community.

2.8 Related Work

2.8.1 Surveys of Sign Language Recognition and Translation

Several surveys have examined computational sign language research, although their primary focus remains on methods and algorithms rather than datasets. Koller (2020) presents a large-scale meta-analysis of sign language recognition, covering approximately 300 published papers and over 400 experimental results from 1983 to 2020. A key finding relevant to this thesis is that a single dataset, RWTH-PHOENIX-Weather, constituted the only large-vocabulary continuous sign language recognition benchmark available worldwide at the time, illustrating the uneven progression of dataset development across task types.

Sarhan and Frintrop (2023) review isolated sign language recognition methods over the preceding decade, addressing modality types, feature extraction approaches, and transfer learning strategies. Although the paper provides an overview of publicly available datasets, it does not offer a structured comparison of dataset characteristics. A subsequent review by Sarhan and Frintrop (2024) expands the scope to include sign language translation and generation, identifying substantial variability across datasets as a central obstacle to developing robust systems, yet again without a dedicated dataset-level analysis. Toshpulatov et al. (2025) present a recent comprehensive review of deep learning approaches to sign language processing, encompassing recognition, translation, and production. Datasets are considered as supporting resources, with the review highlighting gaps in data coverage as a limiting factor for the field.

Collectively, these surveys confirm that dataset characteristics are widely recognized as important, yet no existing survey treats sign language datasets as the primary object of structured analysis.

2.8.2 Studies Directly Using Sign Language Datasets

Several influential papers focus primarily on introducing or utilizing specific sign language datasets, thereby demonstrating the research community's reliance on well-constructed datasets. Li et al. (2020) introduce WLASL, a large-scale American Sign Language dataset containing 2,000 sign classes and recordings from 119 signers, and show that scale and signer diversity significantly influence recognition performance. Sincan and Keles (2020) present AUTSL, a multimodal Turkish Sign Language dataset with both RGB and depth modalities recorded from 43 signers, and provide baseline benchmark results, serving as a concrete example of the benchmarking practices discussed in Section 2.6.

Camgöz et al. (2020) utilize the PHOENIX-2014T dataset to develop a transformer-based model capable of performing both continuous sign recognition and sign language translation within a single framework, demonstrating that annotation depth, specifically the availability of both gloss labels and sentence translations, directly enables more advanced research tasks. Duarte et al. (2021) introduce How2Sign, a large-scale continuous ASL dataset containing approximately 16,000 signed sentences across multiple modalities, illustrating the scale and diversity necessary to support real-world translation research.

Bragg et al. (2021) adopt a different perspective, examining sign language AI datasets through the lens of Fairness, Accountability, Transparency, and Ethics (FATE). Their analysis reveals that all existing datasets include only a small number of signers, demographic information is rarely reported, and the rights and responsibilities of data contributors are insufficiently addressed. This work most closely aligns with the present thesis in that it subjects sign language datasets, rather than the models trained on them, to critical scrutiny.

2.8.3 Systematic Mapping Methodology

The methodology underpinning this thesis draws on established guidelines for systematic reviews and systematic mapping studies in software engineering. Kitchenham and Charters (2007) provide foundational guidelines for conducting systematic literature reviews, adapted from medical research practices to address the specific challenges of software engineering. These guidelines cover the three phases of a review: planning, conducting, and reporting.

Petersen et al. (2008) introduce systematic mapping studies as a distinct approach from systematic literature reviews. While a systematic review aims to answer specific research questions through in-depth analysis of selected studies, a mapping study aims to produce a broad overview of a research area by classifying publications according to predefined attributes and identifying patterns, frequencies, and gaps. This distinction is updated and clarified in Petersen et al. (2015), which provides revised guidelines and emphasises that systematic mapping is particularly appropriate when a field is large and heterogeneous — precisely the conditions that characterise the sign language dataset landscape.

2.8.4 Positioning This Thesis

The related work reviewed above identifies a persistent gap: although sign language datasets are widely used and their limitations frequently acknowledged, no existing study provides a structured, systematic mapping of dataset characteristics across the field. Surveys treat datasets as supporting material, dataset papers emphasize individual contributions, and ethics-oriented analyses such as Bragg et al. (2021), while valuable, do not employ a systematic mapping methodology. This thesis addresses that gap directly by applying the guidelines of Petersen et al. (2008, 2015) and Kitchenham and Charters (2007) to produce a structured overview of sign language datasets classified by language coverage, modality, scale, annotation, and accessibility.

3. Methodology

This chapter outlines the methodology used to identify, select, and analyze sign language datasets. The approach is based on the systematic mapping method proposed by Petersen et al. (2008, 2015) and is supplemented by the systematic review guidelines of Kitchenham and Charters (2007).

3.1 The Systematic Mapping Method (Petersen et al.)

A systematic mapping study is a form of secondary research that produces a structured overview of a research area by classifying and categorizing existing publications according to predefined attributes. The method was introduced by Petersen et al. (2008) and refined in Petersen et al. (2015), drawing on the broader systematic review tradition established in medical research and adapted for software engineering by Kitchenham and Charters (2007).

The primary aim of a systematic mapping study is to establish the structure of a research field, rather than to synthesize findings or address a focused empirical question, which is the purpose of a systematic literature review. Systematic mapping identifies which topics have been studied, their frequency, the methods used, and existing research gaps. Petersen et al. (2008) describe the output as a visual or tabular "map" that distributes publications across classification categories derived from the research questions.

3.2.1 Systematic Mapping vs. Systematic Literature Review

Petersen et al. (2008) explicitly distinguish between systematic mapping studies and systematic literature reviews, as these approaches are frequently conflated. A systematic literature review (SLR) is a detailed and focused investigation that aggregates and synthesizes evidence to answer specific research questions, such as "which machine learning algorithm achieves the highest accuracy for isolated sign recognition?" In contrast, a systematic mapping study adopts a broader, classification-oriented perspective, addressing questions such as "what types of datasets exist, and how are they distributed across languages, modalities, and tasks?"

According to Petersen et al. (2015), systematic mapping is most suitable when the research field is large and heterogeneous, when the objective is to identify trends and gaps rather than synthesize quantitative evidence, and when no structured overview currently exists. All three conditions are met in this study: the sign language dataset landscape encompasses numerous languages, modalities, and task types; the primary aim is classification and gap identification rather than performance synthesis.

3.2.2 The Six-Step Process

Petersen et al. (2008) define systematic mapping as a six-step process. Each step is described below.

Step 1 — Define research questions. The process begins with the formulation of research questions that determine the scope of the mapping and guide the classification scheme. In contrast to the typically narrow and empirical questions of SLRs, mapping research questions are broad and classificatory, for example, "what types of X exist?" or "how is the field distributed across Y?" Petersen et al. (2008) recommend that research questions should correspond directly to the classification categories used in the subsequent data extraction step.

Step 2 — Conduct the systematic search. A keyword-based search is performed across multiple academic databases to ensure comprehensive coverage. Petersen et al. (2008) recommend combining several search terms with Boolean operators and searching at least two to three major databases. The search process should be documented in sufficient detail to enable replication. Backward snowballing, which involves examining reference lists of included papers to identify additional relevant sources, is also recommended as a complementary strategy.

Step 3 — Screen publications. All publications retrieved through the search are screened using predefined inclusion and exclusion criteria. Petersen et al. (2008) specify a two-stage screening process: an initial screen based on title and abstract, followed by a full-text review of the remaining records. The same set of criteria must be applied at both stages, and these criteria should be defined prior to screening to prevent post-hoc adjustments that could introduce bias. Kitchenham and Charters (2007) further emphasize the importance of transparently reporting the number of records retained and excluded at each stage.

Step 4 — Extract data using a predefined keywording scheme. In contrast to systematic reviews, which extract quantitative results, systematic mapping studies focus on extracting classification-relevant metadata from each included publication. Petersen et al. (2008) refer to this process as "keywording," which involves identifying the attributes of each study that are relevant to the research questions. These attributes form the basis of the classification map. A structured extraction form or spreadsheet is used to ensure consistent capture of attributes across all included studies.

Step 5 — Classify and map the results. The extracted data are used to classify publications, or in this study, datasets, across the dimensions identified in Step 1. Petersen et al. (2008) present classification results as frequency counts and visual maps that illustrate the distribution of studies across categories. These maps reveal the structure of the research area and facilitate the identification of patterns and concentrations.

Step 6 — Report findings. The final step is to report the classification results, interpret the patterns identified in the map, and explicitly identify research gaps, which are areas that are underrepresented or absent. Petersen et al. (2015) updated their guidelines to emphasize the importance of discussing both the existing literature and the gaps, as gap identification is a primary contribution of a mapping study.

3.2.3 Validity and Rigour in Systematic Mapping

Petersen et al. (2008) identify several potential threats to the validity of a systematic mapping study. Search incompleteness — the risk that relevant publications are missed due to the choice of databases or search terms — is the most commonly cited threat. They recommend mitigating this through the use of multiple data sources, iterative search refinement, and snowballing. Classification bias — the risk that subjective judgements during categorization introduce inconsistency — is addressed by defining classification categories before extraction begins and by documenting ambiguous cases explicitly rather than resolving them through inference. Both threats and the mitigation strategies applied in this study are discussed in Section 3.8.

3.3 Rationale for Choosing Systematic Mapping

The decision to apply a systematic mapping methodology, instead of a narrative review or a systematic literature review, was guided by the characteristics of the research area and the objectives of this study.

A narrative review is not appropriate in this context because it lacks a structured search process and predefined selection criteria, which increases susceptibility to selection bias and limits replicability. A systematic literature review would necessitate a narrower research question and would aim to synthesize performance results across datasets. However, this is currently unfeasible due to substantial differences in evaluation protocols, metrics, and sign languages across studies, which preclude meaningful quantitative synthesis.

Systematic mapping is the most appropriate approach because the field is large and heterogeneous, no prior structured overview exists, and the primary contribution of this work is the classification of the dataset landscape. As noted by Petersen et al. (2015), mapping is particularly well suited to fields where understanding the scope and structure of existing work is necessary before more focused investigations can be undertaken. This situation applies to sign language dataset research, where the lack of a structured overview has been identified as a barrier to progress.

3.4 Research Questions

Following Step 1 of the Petersen et al. (2008) process, the research questions for this study were formulated to be broad and classificatory, aligned with the mapping objective. The following overarching question and sub-questions were established:

Research Question: How are publicly available sign language datasets designed, and what research directions do their characteristics enable or constrain?

> **Sub-question 1:** What structural and technical characteristics define existing sign language datasets (e.g., language coverage, modality, scale, annotation type)?

Sub-question 2:: What types of computational tasks (e.g., isolated recognition, continuous recognition, translation, sign spotting) are supported by these datasets?

Sub-question 3:: What patterns, limitations, and research gaps emerge from the current dataset landscape?

These questions correspond directly to the classification dimensions used in the extraction scheme (Section 3.6) and to the structure of the results presented in Chapter 4. RQ1 and RQ2 guide the classification categories; RQ3 guides the gap analysis.

3.5 Review Protocol

Consistent with the guidelines of Petersen et al. (2008) and Kitchenham and Charters (2007), a review protocol was established prior to the commencement of the search and screening process. This protocol specifies the data sources, search strategy, selection criteria, and data extraction scheme, and is documented here to ensure transparency and reproducibility.

3.5.1 Data Sources

The search was conducted across four primary academic databases: Google Scholar, IEEE Xplore, ACM Digital Library, and arXiv. These databases were selected for their comprehensive coverage of computer science and computer vision research, which are the main disciplines in which sign language datasets are developed and published. Additionally, project websites, institutional dataset repositories, and research group pages were searched manually to identify datasets documented outside traditional academic venues.

3.5.2 Search Strategy

A keyword-based search strategy was used, combining terms related to sign language and dataset collection. The following representative Boolean search string was applied:

"sign language" AND "dataset" AND ("corpus" OR "recognition") AND NOT "gesture"

Variations of this search string were applied across different databases to accommodate differences in indexing and search syntax. The search was supplemented by backward snowballing, in which reference lists of all included papers were examined to identify additional relevant datasets. Snowballing was concluded when five consecutive papers yielded no new eligible datasets, a saturation criterion recommended by Petersen et al. (2008).

3.5.3 Inclusion and Exclusion Criteria

The following criteria were defined in the protocol before screening began, in accordance with the requirements of Petersen et al. (2008) and Kitchenham and Charters (2007).

Inclusion criteria — a dataset was included if it:

• described a dataset related to a natural sign language (i.e., a language used by a deaf or hard-of-hearing community, not a designed gesture vocabulary);

• provided visual data in the form of video or image recordings;

• was documented in a peer-reviewed paper, conference proceeding, or official dataset description; and

• was documented in English.

Exclusion criteria — a dataset was excluded if it:

• focused solely on gesture recognition unrelated to any natural sign language;

• lacked sufficient documentation to extract the required attributes;

• consisted solely of synthetic or computer-generated signing data; or

• was not publicly accessible or identifiable through public documentation.

3.5.4 Study Selection Process

The selection process followed the two-stage procedure defined by Petersen et al. (2008). In the first stage, titles and abstracts of all retrieved records were screened to exclude clearly irrelevant publications. In the second stage, the full texts of the remaining records were reviewed to confirm eligibility against all inclusion and exclusion criteria.

3.6 Data Extraction

In accordance with Step 4 of the Petersen et al. (2008) process, a predefined extraction scheme was developed to capture the attributes of each included dataset relevant to the research questions. This scheme was applied uniformly to all nine datasets, and the extracted data were recorded in a structured spreadsheet. Where documentation was ambiguous or absent, the value was recorded as 'not specified' rather than inferred, to preserve accuracy and avoid introducing bias.

The following attributes were extracted for each dataset:

• Dataset name, publication year, and full citation.

• Sign language and country or region of origin.

• Dataset type: isolated signs, continuous signing, fingerspelling, or mixed.

• Scale: number of sign classes, number of video samples, number of signers, and total recording duration.

• Modalities: RGB video, depth video, skeleton or pose data, multi-view recording, and audio.

• Annotation depth: gloss labels, sentence-level alignment, and non-manual signal annotation.

• Benchmark practices: availability of a signer-independent evaluation split and published baseline model results.

• Accessibility: public availability, license type, and documentation quality.

• Demographic information: deaf/hearing composition of signers, gender distribution, and age range.

3.7 Classification and Mapping

Following Step 5 of the Petersen et al. (2008) process, the extracted attributes were used to classify datasets across the following six dimensions, which correspond directly to the research questions formulated in Section 3.4:

• Language coverage and geographic distribution (RQ1, RQ3).

• Dataset type and supported computational tasks (RQ1, RQ2).

• Modality and recording setup (RQ1).

• Annotation depth and structure (RQ1, RQ2).

• Benchmark rigour: signer-independent splits and baseline results (RQ1, RQ3).

• Demographic reporting and signer diversity (RQ1, RQ3).

Classification results are presented in Chapter 4 using descriptive tables, consistent with the outputs of systematic mapping studies as described by Petersen et al. (2008). These tables serve as the systematic map of the field, enabling the identification of patterns, concentrations, and gaps as required by RQ3.

3.8 Validity Considerations

Petersen et al. (2008) identify several threats to validity that are common in systematic mapping studies, and Kitchenham and Charters (2007) provide corresponding mitigation strategies. The following threats and mitigations apply to this study:

Search incompleteness. Relevant datasets may be missing from the selected databases, documented only in non-English sources, or distributed through informal channels. To mitigate this, four databases were searched in parallel, manual website searches were conducted, and backward snowballing was applied until saturation.

Selection bias. Subjective judgements during title/abstract screening may introduce inconsistency. Mitigation: inclusion and exclusion criteria were defined in the protocol before screening began and applied uniformly across both screening stages.

Extraction inconsistency. Ambiguous or incomplete dataset documentation may lead to inconsistent attribute recording. Mitigation: ambiguous values were recorded as 'not specified' rather than estimated, and the same extraction scheme was applied to every dataset.

Classification subjectivity. The assignment of datasets to classification categories involves judgement. Mitigation: classification dimensions were defined before extraction began and are directly derived from the research questions, reducing the scope for post-hoc adjustment.

These mitigation strategies align with the recommendations of Petersen et al. (2008, 2015) and Kitchenham and Charters (2007), and are documented here to enhance the reproducibility and transparency of this study.

4. Results and Initial Discussion

This chapter presents the results of the systematic mapping study described in Chapter 3. Applying the Petersen et al. (2008) framework, nine sign language datasets were identified, selected, and analysed according to the predefined extraction scheme. The findings are organised around the six classification dimensions introduced in Section 3.7: language coverage, dataset type, scale, modality, annotation depth, and benchmark and accessibility practices.

4.1 Overview of Included Datasets

A total of seven sign language datasets were identified through the systematic search and selection process outlined in Chapter 3. These datasets were chosen to ensure coverage across various sign languages, dataset sizes, and design features. The selected datasets cover sign languages from several geographic regions and differ in dataset type, modality, annotation detail, and accessibility. Each dataset is described in peer-reviewed publications and includes sufficient metadata for structured analysis. This chapter reports the results of dataset characterization according to the predefined extraction criteria. The findings are structured around key dimensions relevant to dataset usability, accessibility, and research applicability.

4.2 Distribution by Sign Language and Region

The seven datasets cover a diverse range of sign languages, including national sign languages from Europe, South America, Asia, and North America. Each dataset targets a single sign language, reflecting the language-specific approach to sign language data collection. Despite broad geographic representation, the distribution is uneven. Most originate from regions with established research infrastructure in computer vision and machine learning. Several regions and sign languages are underrepresented, indicating gaps in dataset availability. These findings highlight the fragmented development of sign language datasets, which are typically created within isolated national or institutional contexts rather than through coordinated international initiatives.

4.3 Dataset Type and Task Support

Analysis of dataset types indicates that most included datasets focus on isolated sign recognition**.** These datasets generally consist of short video clips, each representing a single sign. Few datasets support continuous signing**,** which involves sequences of signs forming sentences or phrases. Continuous datasets are more complex to collect and annotate, which may explain their lower prevalence. As a result, current publicly available datasets are more suitable for isolated sign recognition tasks than for advanced applications such as sign language translation or discourse-level analysis.

4.4 Dataset Scale and Composition

The included datasets vary substantially in scale. Some contain a limited number of sign classes and samples, while others include hundreds of signs and tens of thousands of video instances. The number of signers also varies considerably across datasets. Larger datasets typically include more signers, supporting greater variation in signing style and appearance. However, detailed demographic information, such as age range or linguistic background, is frequently absent or inconsistently reported. Variation in scale and documentation affects direct comparison between datasets and restricts the assessment of signer diversity across studies.

4.5 Modalities and Recording Setup

Most datasets utilize RGB video as the primary modality. Several datasets also provide additional modalities, such as depth data or skeletal pose information, typically captured with depth sensors. Multimodal datasets provide richer representations but are less common and generally require specialized hardware. In contrast, RGB-only datasets are more prevalent and easier to collect, though they may offer limited detail for fine-grained sign analysis. Recording environments range from controlled studio settings with simple backgrounds to more complex environments with varied backgrounds and lighting. Controlled setups are predominant in isolated-sign datasets, whereas continuous datasets typically exhibit greater environmental variability.

4.6 Annotation and Benchmarking Practices

All included datasets offer some form of annotation, most commonly gloss-level labels for individual signs. However, annotation depth and structure differ considerably across datasets. Only a subset of datasets explicitly defines signer-independent evaluation protocols, in which test data consists of signers not present during training. These protocols are essential for evaluating model generalizability but are not consistently implemented. Not all datasets provide baseline model results. When included, baseline evaluations vary in model architecture, evaluation metrics, and reporting standards, which limits direct performance comparisons across datasets.

4.7 Accessibility and Availability

All datasets included in this initial analysis are described as publicly available or accessible upon request for research purposes. However, licensing information is often incomplete or vaguely specified. Documentation quality also varies. Some datasets include comprehensive descriptions, download instructions, and evaluation protocols, whereas others provide minimal guidance beyond the associated publication. These differences affect dataset accessibility and practical usability, especially for researchers who are new to the field.

4.8 Initial Discussion

The initial mapping indicates a strong emphasis on isolated sign recognition datasets and controlled recording environments. While this supports early-stage model development, it restricts the evaluation of model generalization and real-world applicability. The lack of consistent signer-independent benchmarks and incomplete demographic reporting further complicates cross-dataset comparisons. These findings indicate a need for improved dataset documentation practices and standardized evaluation criteria, which will be explored further as the dataset corpus expands.

5. Conclusion

This thesis presents a systematic mapping and initial benchmarking of sign language datasets, offering a structured overview of their key characteristics and current limitations. By identifying and categorizing datasets based on language coverage, modality, scale, and annotation, this work establishes a clear baseline for understanding the current landscape of sign language research datasets. The initial findings indicate that many available datasets primarily support isolated sign recognition and are collected in controlled environments. Although these datasets are valuable for early-stage research, they provide limited support for evaluating generalization and real-world applicability. Furthermore, inconsistencies in dataset documentation and evaluation practices hinder direct comparison across datasets. The mapping and benchmarking framework proposed in this thesis serves as a foundation that can be extended as additional datasets are incorporated. Future research can build on this baseline by expanding the dataset corpus, refining evaluation criteria, and examining inclusivity and accessibility in greater detail. Overall, this study provides a transparent starting point for more systematic analysis and informed utilization of sign language datasets.

References

1. Choudhury, T. S., Das, A. K., & Dutta, S. (2022). *Survey on sign language recognition in the context of vision.* Array, 13, 100106. https://doi.org/10.1016/j.measen.2022.100385
2. Koller, O. (2020). *Quantitative survey of the state of the art in sign language recognition* (arXiv:2008.09918). arXiv. https://arxiv.org/abs/2008.09918
3. Ronchetti, F., Quiroga, F., Estrebou, C. A., Lanzarini, L. C., & Rosete, A. (2016). LSA64: An Argentinian sign language dataset. In *Proceedings of the 22nd Congreso Argentino de Ciencias de la Computación (CACIC)* (pp. 191–200). https://doi.org/10.48550/arXiv.2310.17429
4. Sincan, O. M., & Keles, H. Y. (2020). AUTSL: A large scale multi-modal Turkish Sign Language dataset and baseline methods. *IEEE Access, 8*, 181340–181355. https://doi.org/10.48550/arXiv.2008.00932
5. Duarte, A., Palaskar, S., Ventura, L., Ghadiyaram, D., DeHaan, K., Metze, F., Torres, J., & Giro-i-Nieto, X. (2021). How2Sign: A large-scale multimodal dataset for continuous American Sign Language. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 2735–2744). https://doi.org/10.48550/arXiv.2008.08143
6. Salonen, J., Kronqvist, A., & Jantunen, T. (2020). The Corpus of Finnish Sign Language. In *Proceedings of the LREC2020 9th Workshop on the Representation and Processing of Sign Languages: Sign Language Resources in the Service of the Language Community, Technological Challenges and Application Perspectives* (pp. 197–202). European Language Resources Association (ELRA). https://aclanthology.org/2020.signlang-1.32/
7. Rîpanu, C.-A., Hotnog, A.-T., Imbrea, G.-S., & Cercel, D.-C. (2025). RoCoISLR: A Romanian corpus for isolated sign language recognition. *arXiv*. https://doi.org/10.48550/arXiv.2511.12767
8. Rubaiyeat, H. A., Mahmud, H., Habib, A., & Hasan, M. K. (2024). BdSLW60: A word-level Bangla Sign Language dataset. *arXiv*. https://doi.org/10.48550/arXiv.2402.08635
9. Rayeed, S. M., Tamzida Tuba, S., Mahmud, H., Mazumder, M. H. U., Hossain, M. S., & Hasan, M. K. (2023). BdSL47: A complete depth-based Bangla sign alphabet and digit dataset. *Mendeley Data, V3*. https://doi.org/10.17632/pbb3w3f92y.3